

Advanced (Habanero)

Q1. Convert $x^2 + 6x + 8$ to vertex form.

- (A) $(x + 3)^2 - 1$
 (B) $(x + 3)^2 + 1$
 (C) $(x + 4)^2 - 8$
 (D) $(x + 4)^2 + 8$
 (E) $(x - 3)^2 - 1$

Q2. Complete the square for $x^2 + 4x$.

- (A) $(x + 2)^2 - 4$
 (B) $(x + 2)^2 + 4$
 (C) $(x - 2)^2 - 4$
 (D) $(x - 2)^2 + 4$
 (E) $(x + 1)^2 - 1$

Q3. The vertex form of $x^2 - 2x - 3$ is

- (A) $(x - 1)^2 - 4$
 (B) $(x - 1)^2 + 4$
 (C) $(x + 1)^2 - 4$
 (D) $(x + 1)^2 + 4$
 (E) $(x - 2)^2 - 7$

Q4. What is the vertex of the parabola $y = x^2 - 4x - 3$?

- (A) $(2, -7)$
 (B) $(2, -3)$
 (C) $(2, 7)$
 (D) $(-2, -3)$
 (E) $(-2, 7)$

Q5. Complete the square for $x^2 - 6x + 2$.

- (A) $(x - 3)^2 - 7$
 (B) $(x - 3)^2 + 7$
 (C) $(x + 3)^2 - 7$
 (D) $(x + 3)^2 + 7$
 (E) $(x - 2)^2 - 2$

Q6. The equation $y = (x - 2)^2 + 1$ has its vertex at

- (A) $(2, 0)$
 (B) $(2, 1)$
 (C) $(0, 1)$
 (D) $(0, 2)$
 (E) $(-2, 0)$

Q7. What is the standard form of $(x + 2)^2 - 3$?

- (A) $x^2 + 4x + 1$
 (B) $x^2 - 4x + 1$
 (C) $x^2 + 4x - 1$
 (D) $x^2 - 4x - 1$
 (E) $x^2 + 4x + 4$

Q8. The equation $y = x^2 - 8x + 7$ can be rewritten in vertex form as

- (A) $(x - 4)^2 - 9$
 (B) $(x - 4)^2 + 9$
 (C) $(x + 4)^2 - 9$
 (D) $(x + 4)^2 + 9$
 (E) $(x - 2)^2 - 7$

Open Ended Questions (FRQ)**Q9.** Convert $x^2 + 2x - 6$ into vertex form and identify its vertex.**Q10.** Complete the square for $x^2 - 10x - 3$ and then write the equation in standard form.**Chef's Tips**

- Allow students to use a calculator for checks, but not for completing the square.
- Encourage students to verify their vertex form equations by expanding and checking.
- Consider providing examples of real-world applications to help students see the relevance of vertex form.